JAYESH KOLI

Shirpur 425427, M.H. +91 8530379044 jayeshkoli1402@gmail.com
LinkedIn GitHub Portfolio

OBJECTIVE

AI/ML Developer focused on building real-world machine learning projects, with practical experience in data preprocessing, model evaluation, and handling production-level data challenges like class imbalance.

EDUCATION

B.Tech (Computer Science and Engineering)
SVKM's College of Engineering, Daivad — Shirpur
[2024 – 2028] | Pursuing

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Java, JavaScript.
AI/ML: NumPy, Pandas, Matplotlib, Scikit-learn.
Machine Learning: KNN, Linear Regression, Train-Test Split, Random Forest, Logistic Regression, Feature Scaling, Data Preprocessing, Model Evaluation.
Web Backend: HTML, CSS, JavaScript, Flask.
Version Control: GitHub, Git.
Development Tools: VS Code, Replit, Jupyter Notebook.

SOFT SKILLS

- Demonstrated **leadership** in guiding team projects and resolving challenges.
- **Adaptable** and quick to learn new tools, technologies, and environments.

PROJECTS

- **CUSTOMER CHURN PREDICTION**
  - Built an end-to-end churn prediction model using Logistic Regression on real-world telecom data with inconsistent categories, missing values, and class imbalance (~27% churn).
  - Designed a preprocessing pipeline using ColumnTransformer and Pipeline to prevent data leakage between train and test sets.
  - Diagnosed misleading accuracy caused by class imbalance; applied class_weight="balanced", improving churn recall from 0.59 to 0.83.
  - Evaluated using confusion matrix, precision, recall, and F1-score instead of relying on accuracy alone.
  - **Technologies:** Python, Pandas, Scikit-learn (Pipeline, ColumnTransformer, LogisticRegression).
  - **GitHub:** https://github.com/jayeshkoli-1402/Customer-Churn-Prediction
- **DIABETES PREDICTION WEB APP**
  - Built a KNN-based diabetes prediction model using Scikit-learn.
  - Improved prediction accuracy from 25% to 70.78% using feature scaling and k-value optimization.
  - Developed a Flask backend to process prediction requests.
  - Used Pandas, NumPy, and Matplotlib for preprocessing and visualization.
  - **Technologies:** Python, Flask, Scikit-learn, Pandas, NumPy, Matplotlib.
  - **GitHub:** https://github.com/jayeshkoli-1402/diabetes-knn-predictor
- **FRAUD DETECTION MODEL**
  - Built a machine learning model to detect fraudulent credit card transactions using Random Forest Classifier on a heavily imbalanced real-world dataset.
  - Applied techniques to handle class imbalance and evaluated model performance using precision, recall, and F1-score metrics.
  - Demonstrated strong understanding of real-world data challenges including noise, skew, and threshold tuning.
  - **Technologies:** Python, Scikit-learn, Pandas, NumPy, Matplotlib.
  - **GitHub:** https://github.com/jayeshkoli-1402/Fraud-Detection-Model

CERTIFICATIONS

- The Joy of Computing using Python — NPTEL (IIT Madras), Score: 90%.